

What is a SerDes?

A serialiser/deserialiser, or SerDes, is a transceiver that converts parallel data to serial data, processes at high data rates and connects again to parallel interface. It enables transmission of large amount of data from one point to another over a single/differential line, thereby reducing the number of data paths, and hence the number of pins, complexity, cost, power and board space of the circuit.

mandated by various committees like USB.org, PCI-SIG, SATA, IEEE and SerDes.

The innovative design of this SerDes would help the OEMs in reducing the number additional off-the-chip components, such as high-precision resistors, clocking solutions and current-bias circuits. The complete solution is built on monolithic silicon as a single block. Thus the reduced number of external connectors and packaging complexities simplifies the test solution and reduces overall size, cost and power consumption of the system.

This SerDes is a specification-driven product that needs to meet stringent requirements of high-speed interface technologies. Its innovative transistor-level circuit design is key factor for the optimal solution that provides low power, high performance, cost saving for applications that meet customers' expectations. It comes with built-in PLLs designed for 10GHz frequencies for clocking solutions. A non-chip, high-precision bias current generator in the design helps to reduce the necessity for multi-chip modules or single-module solution with clocking, data transfer and multiple bias generators.

Design challenges

It took almost three years to claim the finished SerDes IP design. Currently, there are only a handful of companies across the globe that provide such high-speed SerDes IPs. The product is first of its kind developed completely in India and meets the requirements prescribed by the standards bodies like IEEE, USB.org and PCI-SIG.

The complexity of the design being very high, its development required manpower that had years of experience in designing SerDes IPs and their sub-blocks. Scarcity of highly knowledgeable, skilled and experienced workforce in this field was a challenge.

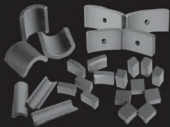
The product has received the 'Most Innovative Product of the Year 2013' award from India Electronic and Semiconductor Association (IESA) in recognition of innovative ideas that were implemented in the design of SerDes, which helped the customers to mitigate their SoC and package complexities.

This is a finished product that can be readily adopted by fabless design houses or OEMs. These products are offered through flexible licensing, royalty or buyout options. It can be sourced to customers worldwide through the web portal Design & Reuse (D&R). The IPs can also be picked up by OEMs who intend to do fab-out of their designs through foundries like Taiwan Semiconductor Manufacturing Company (TSMC), GlobalFoundries and Integrated Device Manufacturer (IDM). ●

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